

III. $\Sigma = LL^T$: cholesky decomposition, L : full column rank

$$y = Lz \sim N_n(0, \Sigma), \quad z \sim N(0, I)$$

$$y^T A y = z^T \underbrace{L^T A L}_B z$$

$$B^T = (L^T A L)^T = L^T A L = B : \text{symmetric}$$

$$L^T A L L^T A L L^T = L^T A L L^T$$

Since L is full column rank, $\exists (L^T L)^{-1}$

$$\cancel{(L^T L)^{-1} L^T L} L^T A L L^T A L L^T = \cancel{(L^T L)^{-1} L^T L} L^T A L L^T$$

$$\Rightarrow L^T A L L^T A L L^T = L^T A L L^T$$

$$\Rightarrow L^T A L L^T A L L^T L^T L \cancel{(L^T L)^{-1}} = L^T A L L^T \cancel{(L^T L)^{-1} L^T L}$$

$$\Rightarrow L^T A L L^T A L = L^T A L$$

$$\Rightarrow L^T A L : \text{idempotent}$$

$$\text{rank}(L^T A L) = \text{tr}(L^T A L) = \text{tr}(A L L^T) = \text{tr}(A \Sigma) = 1$$

$$z^T B z = y^T A y \sim \chi^2_1$$

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$$\begin{aligned}
 (a) \quad \mathbb{E}[y^T W y] &= \mathbb{E}[\text{tr}(y^T W y)] \\
 &= \mathbb{E}[\text{tr}(W y y^T)] \\
 &= \text{tr}(\mathbb{E}[W y y^T]) \\
 &= \text{tr}(W(\Sigma + \mu \mu^T)) \\
 &= \sigma^2 \text{tr}(W) + \text{tr}(\mu^T W \mu) \\
 &= \sigma^2 \text{tr}(W) + \beta^T X^T W X \beta
 \end{aligned}$$

$$(b) \quad u(\beta, \lambda) = (y - X\beta)^T (y - X\beta) + \lambda^T (C\beta - b)$$

$$\frac{\partial u}{\partial \lambda} = C\beta - b = 0 \Rightarrow C\hat{\beta}_n = 0$$

$$\begin{aligned}
 \frac{\partial u}{\partial \beta} &= -2X^T(y - X\beta) + C^T \lambda \\
 &= -2X^T y + 2X^T X \beta + C^T \lambda = 0
 \end{aligned}$$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y - \frac{1}{2} (X^T X)^{-1} C^T \lambda = 0$$

$$C\hat{\beta}_n = C(X^T X)^{-1} X^T y - \frac{1}{2} C(X^T X)^{-1} C^T \lambda = b$$

$$\lambda = 2 [C(X^T X)^{-1} C^T]^{-1} (C(X^T X)^{-1} X^T y - b)$$

$$\begin{aligned}
 \hat{\beta}_n &= \hat{\beta} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C\hat{\beta} \\
 &\quad + (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} b
 \end{aligned}$$

$$= (I - B)\hat{\beta} + c, \text{ where } c = (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} b$$

$$(c) \hat{\beta}_c = (I - B)\hat{\beta} + c$$

$$\text{Cov}((I - B)\hat{\beta} + c, C\hat{\beta})$$

$$= (I - B) \text{var}(\hat{\beta}) C^T$$

$$= \sigma^2 (I - B) (X^T X)^{-1} C^T$$

$$= 0 \quad \because \hat{\beta}_c \perp C\hat{\beta}$$

□

$$(d) \text{var}(\hat{\beta}_c) = \sigma^2 (I - B) (X^T X)^{-1} (I - B)^T$$

$$= \sigma^2 \underbrace{(I - B) (X^T X)^{-1}}_{= Q}$$

$$Q\lambda = 0 \Rightarrow \lambda = 0 \quad \text{invertible}$$

$$\Rightarrow \lambda \neq 0 \quad \text{not invertible}$$

$$\text{Let } y \neq 0, C \neq 0 \quad \lambda := C^T y$$

$$\Rightarrow Q\lambda = 0 \text{ but } \lambda \neq 0 \quad \because \text{rank-deficient.}$$

□

(e)

$$\begin{aligned} C\hat{\beta} - b &= C(X^T X)^{-1} X^T y - b \\ &= C(X^T X)^{-1} X^T [y - X C^T (C C^T)^{-1} b] \end{aligned}$$

$$\begin{aligned} (C\hat{\beta} - b)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - b) \\ = [y - X C^T (C C^T)^{-1} b]^T \underbrace{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C(X^T X)^{-1} X^T}_{= A} [y - X C^T (C C^T)^{-1} b] \end{aligned}$$

$$A^2 = A$$

$$\text{rank}(A) = \text{tr}(A) = \text{tr}(I_q) = q$$

$$\frac{1}{2} \mu^T A \mu = \frac{(C\hat{\beta} - b)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - b)}{2\sigma^2} =: \lambda$$

$$\frac{(C\hat{\beta} - b)^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta} - b)}{\sigma^2} \sim \chi^2(q, \lambda)$$

$$\frac{SSE}{\sigma^2} = \frac{y^T (I - H) y}{\sigma^2}$$

$$= \frac{[y - X C^T (C C^T)^{-1} b]^T (I - H) [y - X C^T (C C^T)^{-1} b]}{\sigma^2}$$

$$\text{rank}(I - H) = \text{tr}(I - H) = n - p - 1$$

$$\frac{1}{2} \mu^T A \mu = 0$$

$$\frac{SSE}{\sigma^2} \sim \chi^2(n-p-1)$$

$$A(I-H) = 0 \quad \text{; independent}$$

$$F = \frac{\chi^2(q, \lambda) / q}{\chi^2(n-p-1) / (n-p-1)} \sim F(q, n-p-1, \lambda)$$

□

$$(f) \quad (\hat{c} - b)^T [C(X^T X)^{-1} C^T I] (\hat{c} - b) \leq q \cdot s^2 \cdot F_{q, n-p-1}(\alpha)$$

$$b \in \{ b \in \mathbb{R}^q : (\hat{c} - b)^T [C(X^T X)^{-1} C^T I] (\hat{c} - b) \leq q \cdot s^2 \cdot F_{q, n-p-1}(\alpha) \}$$

$$\Leftrightarrow \text{reject } H_0 : c\beta = b$$

□

[3].

(a) Let $Q = A^T A (A^T A)^{-1}$, $P = I$

$$Q A^T A = A^T A, \quad P A^T A = A^T A$$

$$0 = (Q A^T A - P A^T A) (Q^T - P^T)$$

$$= (Q A^T - P A^T) (A Q^T - A P^T)$$

$$= B^T B \Rightarrow B = 0$$

$$A Q^T = A P^T$$

$$\Rightarrow A = A (A^T A)^{-1} A^T A$$

(b) $A A^+ A = A \Rightarrow A^T (A^+)^T A^T = A^T \Rightarrow (A^+)^T = (A^T)^+$

$$\Rightarrow (A (A^T A)^{-1} A^T)^T = A ((A^T A)^{-1})^T A^T$$

$$= A (A^T A)^{-1} A^T \quad ; \text{ symmetric}$$

$$\underbrace{A (A^T A)^{-1} A^T A (A^T A)^{-1} A^T}_{= A} = A (A^T A)^{-1} A^T \quad ; \text{ idempotent}$$

Let G_1 and G_2 s.t. $A G_1 A^T = A$,
 $A G_2 A^T = A$

$$0 = (A G_1 A^T - A G_2 A^T) (G_1^T A^T - G_2^T A^T)$$

$$= (A G_1 A^T - A G_2 A^T) (A G_1 A^T - A G_2 A^T)^T$$

$$= B^T B \Rightarrow B = 0 \Rightarrow A G_1 A^T = A G_2 A^T ; \text{ invariant}$$

□.

$$(a) z_{ij} = \log d_{ij} - \log \bar{G}_i, \bar{G}_i = \left(\prod_{j=1}^p d_{ij} \right)^{1/p}$$

$$= \log d_{ij} - \frac{1}{p} \sum_{k=1}^p \log d_{ik}$$

$$\sum z_{ij} = \sum_{j=1}^p \left(\log d_{ij} - \frac{1}{p} \sum_{k=1}^p \log d_{ik} \right)$$

$$= \sum_{j=1}^p \log d_{ij} - p \cdot \frac{1}{p} \sum_{k=1}^p \log d_{ik}$$

$$= \sum_{j=1}^p \log d_{ij} - \sum_{k=1}^p \log d_{ik}$$

$$= 0$$

$\Rightarrow z1 = 0$ (i.e. linearly dependent)

$\Rightarrow \text{rank}(z) < p \Rightarrow \text{not identifiable.}$

□

$$(b) z1 = 0 \Rightarrow N(z) = \text{span}\{1\}$$

$$\sum_{j=1}^p \beta_j = 1^T \beta = 0 \Rightarrow \text{valid.}$$

□

$$(c) \text{ let } C := (1, 1, -1, -1, 0, \dots, 0)$$

$$C^T 1 = 0$$

$$d_{ij} = \frac{\exp(w_{ij})}{\sum_{k=1}^p \exp(w_{ik})}$$

$$\log \pi_{ij} = w_{ij} - \underbrace{\log \left(\frac{1}{K} \sum_{k=1}^K \exp(w_{ik}) \right)}_{= C_i}$$

$$= w_{ij} - C_i$$

$$\begin{aligned} z_{ij} &= (w_{ij} - C_i) - \frac{1}{p} \sum_{k=1}^p (w_{ik} - C_i) \\ &= w_{ij} - \overline{w_{i.}} \end{aligned}$$

$$P := I - \frac{1}{p} J P \Rightarrow Z = W P$$

$$Z^T Z \hat{\beta} = Z^T y$$

$$\begin{aligned} \Rightarrow \hat{\beta} &= (Z^T Z)^{-1} Z^T y \\ &= (P^T W^T W P)^{-1} P^T W^T y \\ &= \frac{1}{n} (P^T P)^{-1} P^T W^T y \\ &= \frac{1}{n} W^T y \end{aligned}$$

$$C \hat{\beta} = \frac{1}{n} C^T W^T y$$

$$\begin{aligned} \Rightarrow \text{var}(C \hat{\beta}) &= \frac{1}{n} C^T W^T \sigma^2 W C \\ &= \frac{\sigma^2}{n} C^T C \\ &= \frac{\sigma^2}{n} 4 \end{aligned}$$

$$SSE = y^T (I - H_E) y$$

$$H_E = Z (Z^T Z)^{-1} Z^T = W P \left(\frac{1}{n} P^T \right) P^T W^T = \frac{W P W^T}{n}$$

$$\text{tr}(H_E) = \text{tr}(P) = p - 1$$

$$\hat{\sigma}^2 = \frac{SSE}{n-p+1} \sim \chi^2(n-p+1) / (n-p+1)$$

$$(Z^T Z)^{-1} Z^T (I - H) z = 0 : \text{indep}$$

$$T = \frac{CT\hat{\beta} - 0}{\sqrt{\frac{\hat{\sigma}^2}{n-4}}} = \frac{CT\hat{\beta}}{\sqrt{\frac{\hat{\sigma}^2}{n-4}}} \perp \sim t(n-p+1)$$

$$\therefore T > t_{\alpha, n-p+1}$$

□